

Store at: +2+8°C.

Presentation:

Cod. TL060 CONT: R1 25 ml / R2 7,5 ml.

Procedure

Reagent for quantitative measurement of Ferritin

Only for in vitro use in clinical laboratory (IVD)

TEST SUMMARY

Latex particles coated with specific anti-human ferritin are agglutinated when mixed with samples containing ferritin. The agglutination causes an absorbance change, dependent upon the ferritin contents of the sample that can be quantified by comparison from a calibrator of known ferritin concentration.

REAGENTS COMPOSITION

Diluent (R1)	Phosphate buffer pH 6.7, containing protein stabilizers. Sodium azide 0.09%
Latex (R2)	Suspension of latex microparticles covalently bound anti-ferritin antibodies suspended in a neutral aqueous solution. Sodium azide 0.09%
Calibrator (Optional)	Human serum: Ferritin. 6 x 1 ml. Concentration is stated on the vial label.
Control (Optional)	Human serum: Ferritin 2 x 1. Concentration is stated on the vial label.

PRECAUTIONS

Components from human origin have been tested and found to be negative for the presence of HBsAg and HCV, and of antibody to HIV (1/2). However, handle cautiously as potentially infectious.

Good laboratory safety practices should be followed when handling laboratory reagents or human samples.

REAGENT PREPARATION AND STABILITY

Reagents: Ready to use. Shake the latex vial gently before dispensing its content into the appropriate plastic vials.

Reagent in the original bottle are stable to the expiration date indicated on the label when capped and stored at 2-8°C. Once opened, the reagent is stable 1 month following the storage instructions.

Do not freeze; frozen reagents could change the functionality of the test.

Ferritin Cal: Ready to use. Once opened, stable for 4 weeks at 2-8°C.

Ferritin Control: Ready to use. Stable until the expiration date on the label when stored tightly closed at 2-8°C and contaminations are prevented during their use.

Signs of reagent deterioration:

- Particles and turbidity indicate contamination or reagents deterioration.

All the reagents of the kit are stable up to the end of the indicated month and year of expiry. Stored at tightly closed at 2-8°C. Do not use reagents over the expiration date.

CALIBRATION

The assay is calibrated against WHO 80/578 International Standard.

The use of other commercially available Ferritin calibrators or controls are not recommended.

SPECIMEN

Specimens should be collected by venipuncture following good laboratory practices. Suitable assay specimens are human serum samples, as fresh as possible (stored up to 7 days at +2...+8°C) or deep-frozen. Any additional clotting or precipitation, which occurs due to the freeze/thaw cycle, should be removed by centrifugation prior to assay.

Very lipemic specimens, or turbid frozen specimens after thawing, must be clarified before the assay by high-speed centrifugation (15 min at approx. 15.000 rpm).

Discard contaminated specimen

MATERIAL REQUIRED BUT NOT PROVIDED

- Thermostatic bath at 37°C.
- Spectrophotometer or photometer thermostatable at 37°C with a 600 nm filter.
- Cuvettes with 1 cm light path.

General laboratory equipment

TEST PROCEDURE

A set of standards is used to calibrate the assay (multipoint calibration mode).

Incubation takes place at 37°C.

An automated procedure may be programmed in most automatic analyzers according the following parameters (**we enclose adaptation COBAS MIRA**):

1. Mix 25 µl of sample (standards, control or serum samples) with 200 µl of R1 Ferritin Buffer and 75 µl of R2 Ferritin Latex. Read Absorbance (A1) at a wavelength at 600 nm.
2. Read Absorbance (A2) at a wavelength at 600 nm after 4 minutes of (A1) reading.

Adaptations for many automatic instruments are available on request.

CALCULATIONS

Calculate the Absorbance increase for individual samples and Blank as follows: **A2 – A1**.

Subtract the Absorbance increase of Blank from Absorbance increase of all the samples. Plot absorbances versus concentration on linear graph paper or use an appropriate curve fitting parameter program.

QUALITY CONTROL

SERUM CONTROLS are recommended to monitor the performance.

Serum controls are recommended for internal quality control. Each laboratory should establish its own Quality Control scheme and corrective actions if controls do not meet the acceptable tolerances.

REFERENCE VALUES

Children and adolescents	15-120 ng/ml.
Men	30-300 ng/ml.
Women (pre-menopausal)	10-160 ng/ml.
Women (post-menopausal)	30-300 ng/ml.

It is suggested that each laboratory establish its own reference range

CLINICAL SIGNIFICANCE

Serum ferritin concentration usually reflects body iron stores and is considered one of the most reliable indicators of iron status of patients. Whereas low serum concentrations of ferritin are always indicative of an iron deficiency, elevated concentrations can occur for variety of reasons. Thus, although elevated concentrations often indicate an excessive iron intake, they are also caused by liver disease, chronic inflammation and malignancies. Pregnant women, blood donors, hemodialysis patients, adolescents and children are groups particularly at risk.

Clinical diagnosis should not be made on a single test result; it should integrate clinical and other laboratory data.

REAGENT PERFORMANCE

- **Linearity:** up to 500 ng/ml ^(note 1).
- **Prozone effect:** No prozone effect was detected up to 5500 ng/mL
- **Detection limit:** Values less than 5.2 ng/mL give non-reproducible results.
- **Sensitivity:** Values less than 5.2 ng/mL
- **Precision:**

	Intra-assay n=10		Inter-assay n=10	
Mean (ng/mL)	182.7	287.2	180.5	286.6
SD (ng/mL)	2.9	4.5	5.9	9.3
CV (%)	1.6	1.6	3.3	3.2

- **Accuracy:** Results obtained using GPL reagents (y) were compared to those obtained using a commercial reagent (x) with similar characteristics. 46 samples ranging from 5 to 500 ng/mL were assayed. The correlation coefficient (r) was 0.995 and the regression equation was $y = 1.01x + 0.34$. The results of the performance characteristics depend on the used analyzer.

INTERFERING SUBSTANCES

Do not interfere:

- Bilirubin: > 20 mg/dL
- Hemoglobin: > 2000 mg/L
- Lipids: > 500 mg/L.

Other substances may interfere⁶.

NOTES

- 1- Samples with higher concentrations should be diluted 1/3 in NaCl 9 g/L. and retested again. The linearity limit depends on the sample / reagent ratio, as well the analyzer used. It will be higher by decreasing the sample volume, although the sensitivity of the test will be proportionally decreased.
- 2- Clinical diagnosis should not be made on findings of a single test result, but should integrate both clinical and laboratory data.

BIBLIOGRAPHY

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GPLTLDTT20-I
Rev. 4 – 29/07/20

Store at: +2+8°C.

Presentation:

Cod. TL060 CONT: R1 25 ml / R2 7,5 ml.

Procedure

FERRITIN TURBI

Procedure for COBAS MIRA PLUS

Preparation of working reagent:

Follow the instructions given on the procedure sheet of reagent.

Calibration:

Use serum calibrator.

Operation conditions:**GENERAL**

Measurement mode: Absorb
 Reaction mode: R-S-SR1
 Calibration mode: LOGIT/LOG5
 Reagent blank: REAG/DIL
 Cleaner: NO
 Wavelength: 600 nm
 Decimal position: 2
 Units: ng/ml

ANALYSIS

Post. Dil. Factor: 10.00
 Conc. factor: NO
 SAMPLE Vol: 25 µL

DILUENT Cycle: 3
 Name: Saline
 Vol: 20 µL

REAGENT Vol: 200 µL
 Cycle: 1

START REAGENT Vol: 75 µL

DILUENT Cycle: 1
 Name: Saline
 Vol: 0.0 µL

CALCULATION

Sample limit: NO
 Reaction Direction: INCREASE
 Check: ON

CALCULATION (continued)

Antigen excess NO
 Conversion factor: 1.00000
 Offset: 0.00000
 Normal range low: NO
 Normal range high: NO
 Number of steps: 1
 Calc. Step A: KINETIC
 Reading first: 3
 Reading last: 12
 Reaction limit: 3.5000 ΔA
 Point: T2

CALIBRATION

Calibr. Interval: ON REQUEST

Reagent blank:
 Reag. Range low: NO High: NO
 Blank range low: NO High: NO

STD POS:**

1: 0.0*** 2: 37.5***
 3: 75.0*** 4: 150.0***
 5: 300.0*** 6: 500.0***
 7: NO 8: NO

REPLICATE: SINGLE
 Deviation:
 Correction std:

(**) Select one position on the standard rack

(***) Standard values on the labels

